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Bootcamp of Mathematics (BOOM) 2025

Regression

Introduction to Predicting Numbers with ML

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Session Objectives

01. Understand **the regression concept**

02. Learn the **standard workflow** (pipeline) for building a ML model.

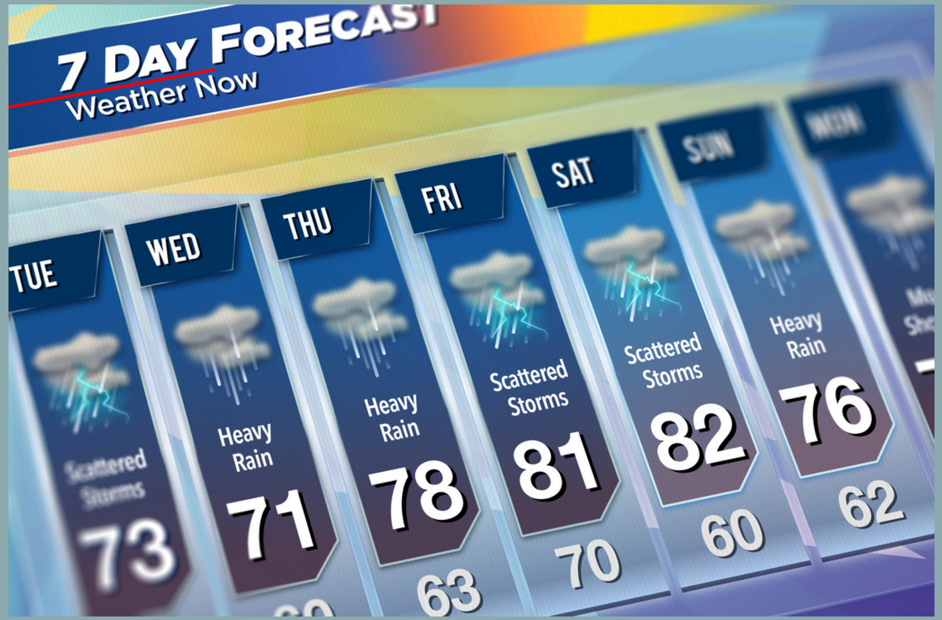
03. Understand **how to evaluate** a model's performance

04. Have **experience building ML regression models** (both industry standard and research standard).

What is Regression?

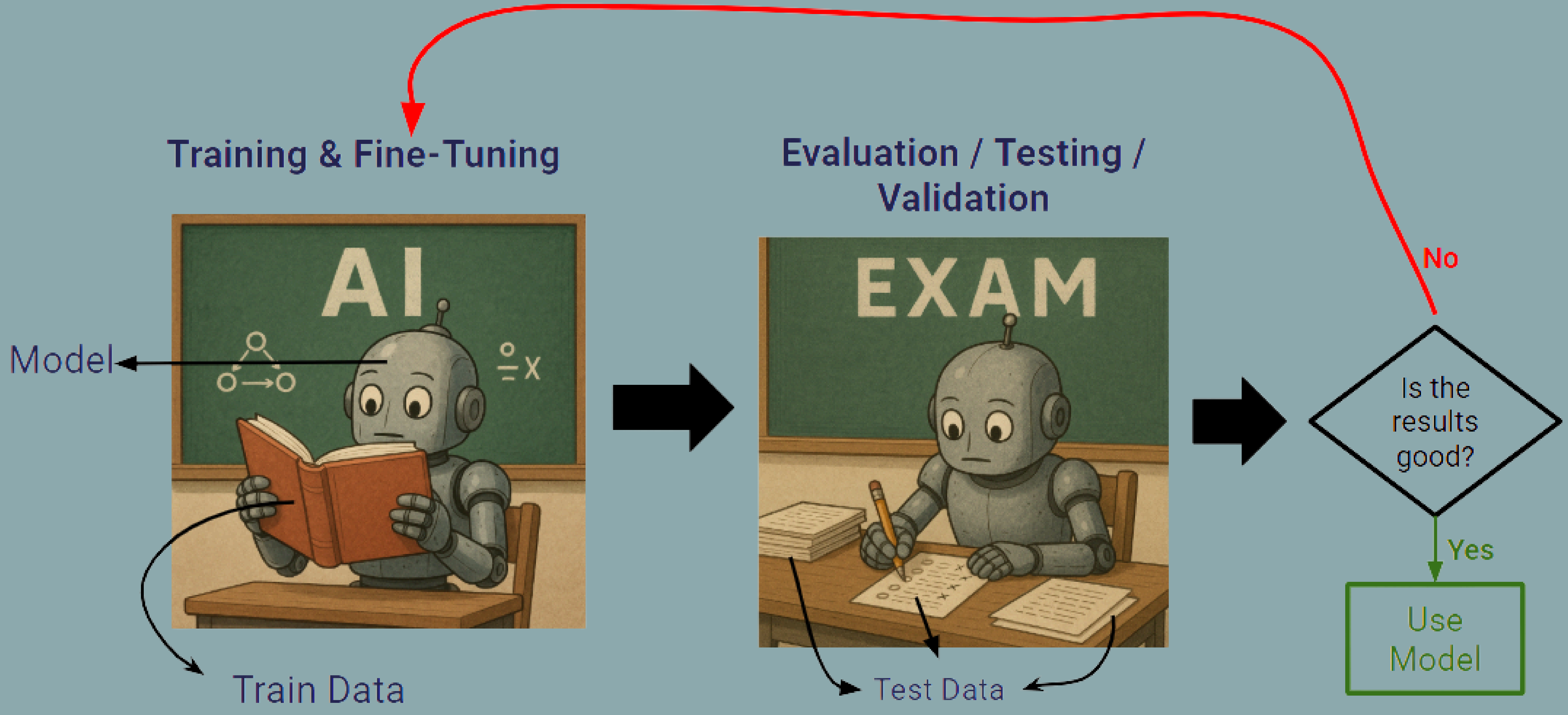
Regression is a type of supervised learning where **the goal is to predict a continuous numerical value**.

- It's about answering "How much?" or "How many?"
- The output is a number on a scale (e.g., 10.5, 1000, 3.14159).



Regression use case: Estimate House Price, Weather Forecasting, Stock Market Prediction

The Machine Learning Workflow



How to Split Data: Hold-Out vs. Cross-Validation (CV)

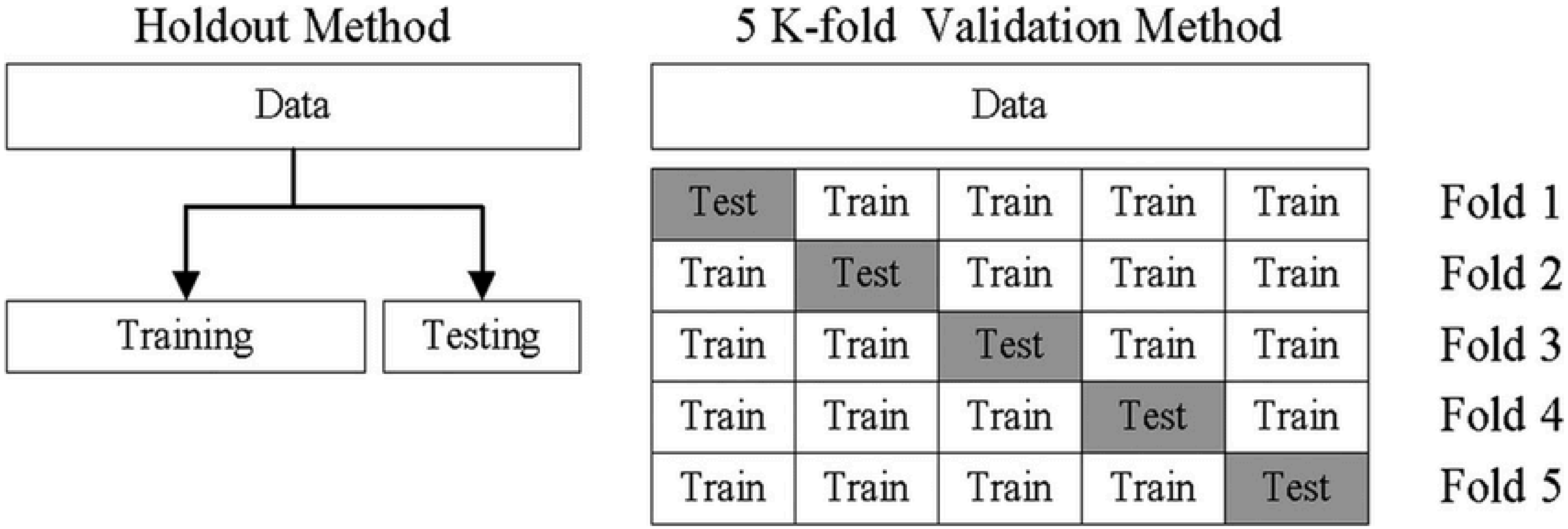
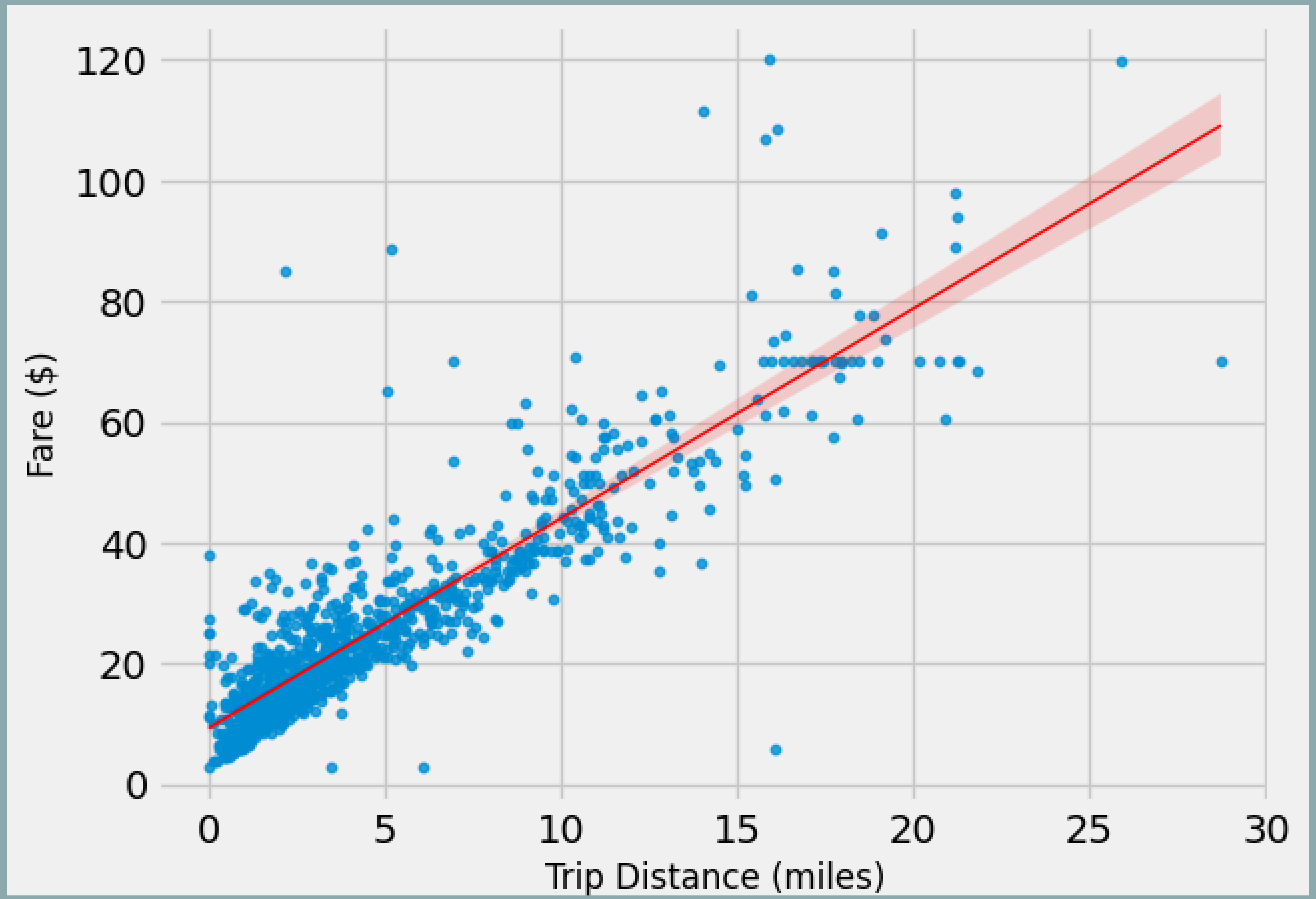


image source: researchgate.com

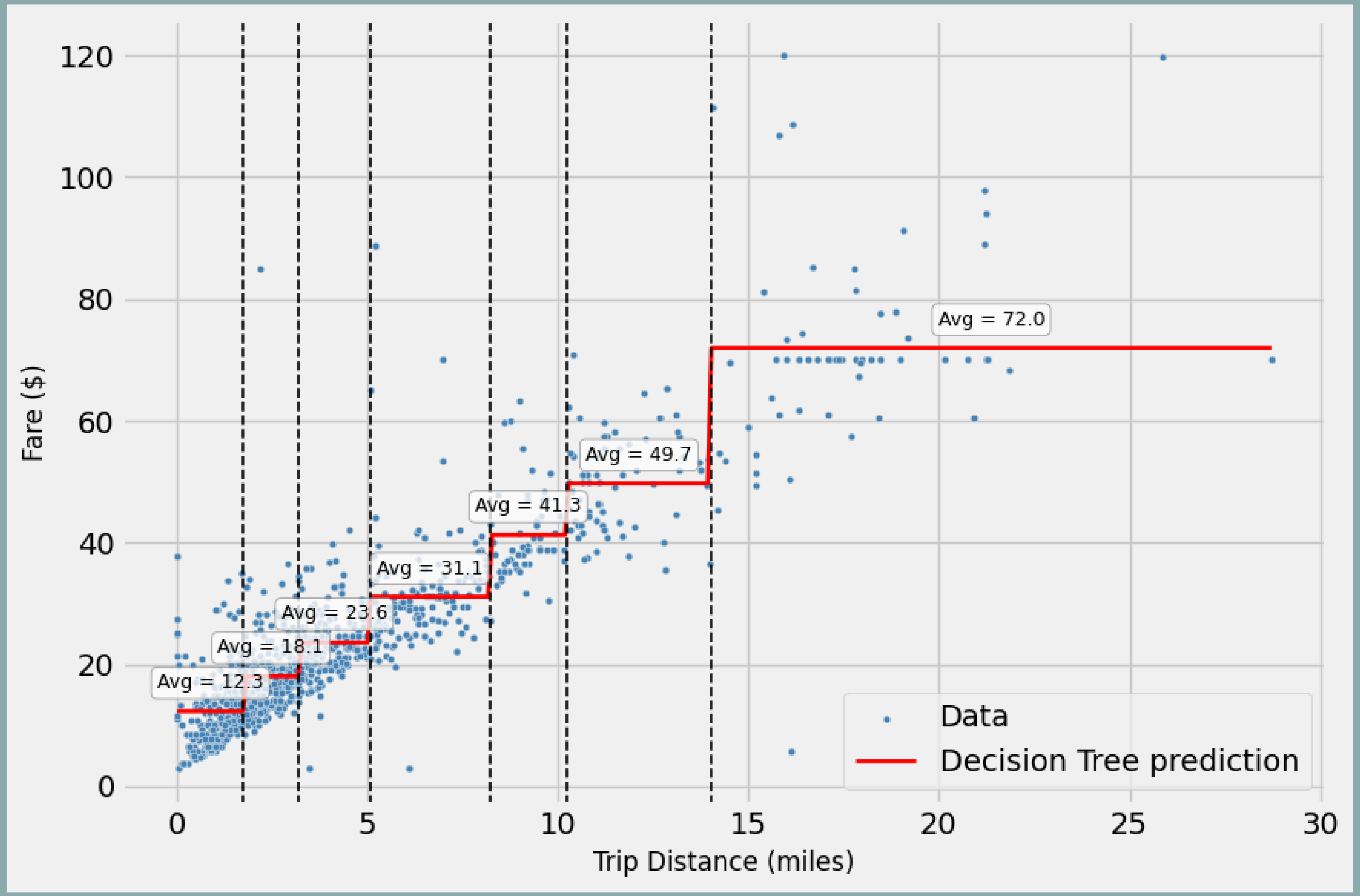
The Basic: Linear Regression

- **Finds the best-fit straight line** through the data points.
- Pros/Cons:
  - **+** : Simple and fast to train.
  - **-** : Assumes a straight-line relationship, which isn't always true.
- Use when the relationship between features and target appears to be linear or as a baseline model.



Decision Trees: Asking Questions to Predict

- Asks a series of yes/no questions to **split the data into groups. The average value of a group is the prediction**.
- Pros/Cons:
  - + : Can model non-linear relationships, very intuitive.
  - - : Can easily become too complex and overfit.
- Use when the relationship between features and target is complex and non-linear.



How Do We Measure Error?

$$\text{MAE}(y, \hat{y}) = \frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}}-1} \left| y_i - \hat{y}_i \right|.$$

- Idea: Measures the average of absolute prediction errors; errors are in the same unit as the target.
- Use/Effect: Good for general regression **when outliers are not extreme**; less sensitive to outliers than MSE/RMSE.

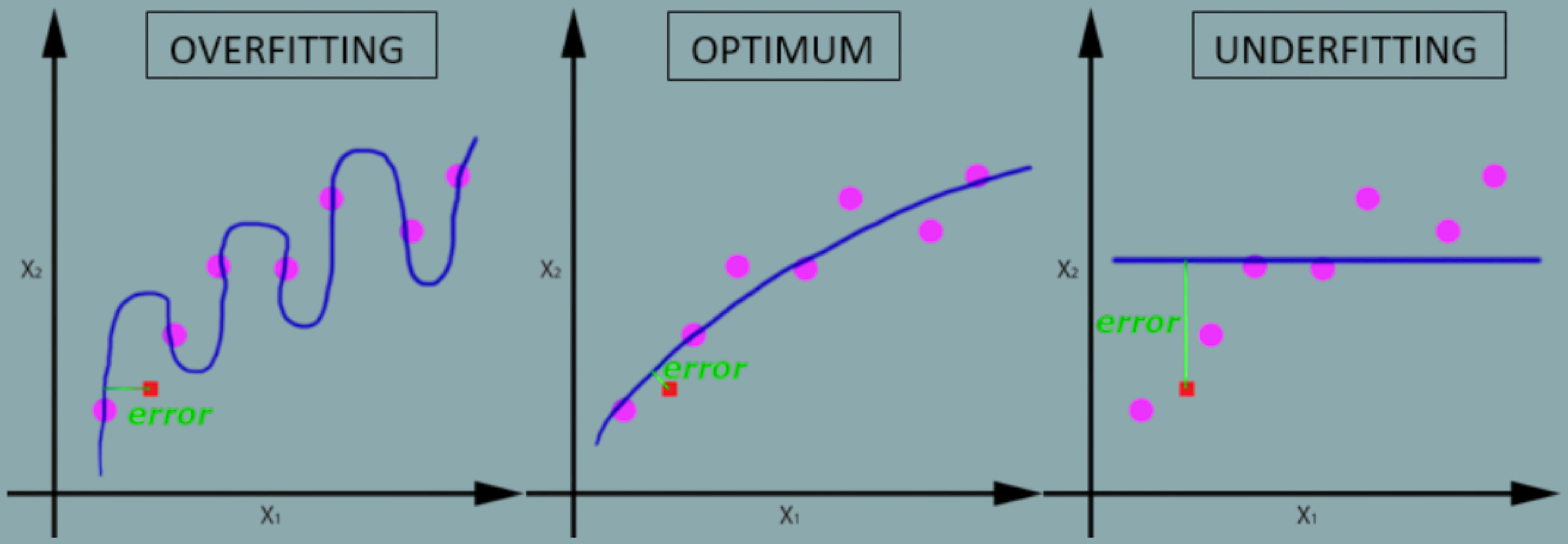
$$\text{MSE}(y, \hat{y}) = \frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}}-1} (y_i - \hat{y}_i)^2.$$

- Idea: Measures the average of squared prediction errors; highly sensitive to large errors (outliers).
- Use/Effect: Best when large errors must be penalized; suitable for linear/polynomial regression with normally distributed errors.

$$\text{MAPE}(y, \hat{y}) = \frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}}-1} \frac{\left| y_i - \hat{y}_i \right|}{\max(\epsilon, \left| y_i \right|)}$$

- Idea: Measures average absolute error as a percentage of actual values; gives a relative error metric.
- Use/Effect: Useful for large-scale, positive-only targets; not reliable when actual values are zero or near zero.

The Balancing Act: Underfitting vs Overfitting



We want the model complexity where the Test Error is lowest (optimum).

Let's Discuss!

Do you have any question?